

Utilizing Transfer Learning on Pre-Trained Models for Image Classification Tasks

Kamini Sharma

*Dept. of Computer Engineering and
Information Technology
P P Savani University
Surat, India
kaminisharma945624@gmail.com*

Pinky Uniyal

*Dept. of Computer Science and
Engineering
Uttaranchal University
Uttaranchal, India
uniyaluniyal24@gmail.com*

Anshu Sharma

*Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, India
asharma406@gmail.com*

Chirag Sharma

*Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, India
chiragsharma1510@gmail.com*

Motashim Rasool

*Dept. of Computer Application
Integral University
Lucknow, India
motashim@iul.ac.in*

Vipin Kumar Chaudhary

*Dept. of Computer Science and
Engineering
Lovely Professional University
Phagwara, India
chaudharyv209@gmail.com*

Abstract—The incredible success of deep learning for image classification due to Convolutional Neural Networks (CNNs) in the past few years. However, training deep networks from scratch requires extensive labelled data and computational resources, so it is not always feasible. Transfer learning, which reuses models trained separately for new but related tasks to accelerate training and enhance accuracy, has emerged as a promising approach for many computer vision tasks. As a result, this paper investigates the transfer learning approach built using different pre-trained CNN architectures, including MobileNetV2, EfficientNetB0, and ResNet50, on image classification (a dataset built from scratch). Tune these models to meet our goals and compare their performance on common evaluation metrics, such as accuracy, Precision, Recall, and F1-score. The study demonstrates that transfer learning accelerates the model's training and can quickly converge to high accuracy with a limited amount of data, making it an effective method in practical image classification tasks when resources are scarce.

Keywords—Artificial Intelligence, Transfer Learning, MobileNetV2, Fine Tuning, Classification

I. INTRODUCTION

Deep learning has had a tremendous impact on numerous fields, including medical image analysis. Convolutional Neural Networks (CNNs) have been shown to be very effective in learning hierarchical image features automatically, which has led to major progress in applications like object recognition, image segmentation or medical diagnostics [2]

All the reactors should be aware of a wonderful technique in deep learning, known as Transfer Learning, which is beneficial for building models of source tasks for target tasks [1]. This is often achieved using transfer learning with pre-trained models, initially trained on large-scale datasets such as ImageNet, as feature extractors. The pre-trained models incorporate a representation learning stage that encodes a wide range of features, including low-level edges

and textures, as well as high-level semantic information, which can be efficiently transferred to downstream tasks.

Yet, transferring learning to a medical image analysis setting poses its own set of challenges. Medical Imaging, such as X-rays, Computed Tomography (CT), and Magnetic Resonance Imaging (MRI), exhibits unique image characteristics that distinguish it from natural images. Often, medical imaging produces greyscale images, which are high-resolution images of complex anatomy or pathology. Thus, applying pretrained models that originate from natural image domains to medical images may not be expected to perform as well.

At its most basic form, classification of images - a rudimentary task that permeates computer scientist Shawni Dang's area of computer vision as one function of understanding visual images - is the task of assigning given images, based on visual content, to a (predefined) category. In this way, image classification is crucial for understanding various applications, ranging from autonomous vehicles' understanding of road scenes to medical professionals' identification of anomalies or diseases from radiological scans [1]. Early architectures relied heavily on handcrafted features (i.e., SIFT and HOG) and concatenated them with some machine learning algorithm (i.e., Support Vector Machines) to classify images [2]. With deep learning, more specifically Convolutional Neural Networks (CNNs), eclipsing and surpassing every known benchmark of state-of-the-art performance for nearly every high-level image recognition application in recent years, the bar has been set considerably high to achieve the same levels of performance [3]. Deep learning models, which are popular for their ability to capture complex patterns from data, have been utilised in various tasks, such as classification and segmentation, by making them more accurate with larger and deeper networks [4]. Convolutional Neural Networks have demonstrated considerable success in capturing informative features from images, resulting in high classification accuracy [5]. Supervised deep learning

approaches for image classification rely on a sufficient amount of well-labelled training data to achieve high accuracy and transferability [6].

A. Background on Image Classification

Image classification involves assigning a label (description) from pre-defined categories to an image, which usually requires a learning process in which an algorithm learns complex patterns and features expected in visual data. Deep Convolutional Neural Networks (deep CNNs) have rapidly become the model of choice for image classification, primarily due to their ability to automatically learn hierarchical image features from raw pixel data. CNN ARCHITECTURAL HIERARCHY 2 CNNs are usually composed of several layers, e.g., a convolutional layer that extracts local features and a pooling layer that reduces dimensionality at the image size, and fully connected perceptrons to classify based on learned features [7]. CNNs rely on their capacity to learn certain patterns from massive amounts of labelled data, hence achieving state-of-the-art performance (a deeper network can understand higher-level features). The popularity of training CNNs owes to their modular design, where layers serve as building blocks, composed of generic layer types that can be utilised in various neural networks [8].

B. Limitations of Training Deep Neural Networks

Training deep neural networks is cumbersome and expensive, as it requires large amounts of labelled data and computational resources. Deep learning methods, especially CNNs, demand a large amount of annotated data to capture the intricacies of shapes and structures in order to obtain a high classification performance on image data; thus, low levels of labelled images can result in what is known as over-fitting — i.e., low performance on unseen data masses while maintaining a fine proportion on seen data. In addition, training deep networks from scratch requires significant computational resources; often, specialised hardware such as GPUs or TPUs is needed, and the process can take days to weeks to converge — making it unfeasible for many real-world applications [9]. One of the major drawbacks, which is a bottleneck to using deep learning models, is the scarce availability of large datasets. Creating such datasets takes many resources, time and effort.

C. Motivation for using Transfer Learning

Transfer learning is a technique used to compensate for the fact that training deep networks from scratch can be impractical or at least slow due to a lack of data by taking advantage of what has been learned in pre-training on large datasets. Transfer learning utilises pre-trained models, which do not require a large amount of labelled data or computational resources; thus, faster training and higher performance are achieved (especially when dealing with constrained data where specialised datasets are unavailable). Pre-trained models trained on large datasets, such as ImageNet, have already learned to encode a very effective feature vector for images, which can be easily adapted to new tasks using transfer learning with minimal fine-tuning.

Transfer learning has the greatest impact on specialised domains where it is difficult to obtain large, labelled datasets, such as in medical imaging or agriculture. It extends the learning from one machine learning task to a related but easier or distinctly different data and domain. For example, if you perform object detection with an extensive Convolutional Neural Network as part of a particular dataset D1 that also covers other tasks such as Semantic Segmentation, then Transfer Learning would use these weights for this part when applied on a similar/related problem; else, using them entirely is not suitable while covering an exclusively new concept. This is when transfer learning becomes a valuable tool in building and deploying powerful machine learning models without requiring large amounts of training data [3].

II. LITERATURE STUDY

Transfer Learning is considered an advanced technique in machine learning, which allows the model to use data samples from one or more domains (source domains) either directly or along with some additional input data of a new domain (target domain). Simply put, in the early days of transfer learning, features learned from a source domain are transferred to a target domain. Deep transfer learning, utilising deep neural networks, has become extremely popular, and pre-trained models such as AlexNet, VGGNet, and ResNet have been widely adopted either as feature extractors or bases for fine-tuning.

Transfer learning has been explored in several domains (applications) such as medical imaging, agriculture and industrial inspection. Transfer learning has been applied to the classification of diseases from X-ray images, CT scans, and MRI images in medical imaging, which helps improve diagnostic accuracy and efficiency [10] [1]. Since it has been proven that transfer learning can accelerate medical imaging classification, as it is too complex to annotate a huge labelling dataset to fully retrain the model [11], we adopt this method. Agriculture: In agriculture, transfer learning has been utilised to detect plant diseases, classify crop types, and monitor crop health using aerial imagery for precision agriculture practices. Transfer learning is applied in industrial inspection to identify defects in manufactured products, grade the quality of these products, and automate visual inspection tasks, thereby enhancing quality control and reducing production costs [14].

III. EXPERIMENTAL SETUP

A. Overview of Pretrained CNN Models

A group of pre-trained CNN models has become de facto in transfer learning; each naturally comes with its own architecture, which leads to certain trade-offs in terms of performance. Skip connections in ResNet models are used to counteract the vanishing gradient problem, allowing for the training of very deep networks. MobileNetV2 is designed for mobile and embedded contexts, offering a lightweight architecture with high performance. In doing so, these pre-trained models offer excellent warm-starting points for transfer learning tasks where the learned features from the auxiliary data can be suitably adapted to specific

tasks [12]. Similarly, we train the ResNet-50 and EfficientNet-B0 models.

B. Methodology

The publicly available dataset used is the MNIST dataset. We split the dataset into an 80:20 ratio. Data augmentation involved applying a series of random transformations to existing images (e.g., rotations, flips, and zooms) to create new synthetic samples. This process enables the model to improve its generalisation and reduces the likelihood of overfitting.

Here, the transfer learning strategy can take one of two forms: feature extraction or fine-tuning. In feature extraction, the pre-trained model acts as a fixed feature extractor where the output of one (or more) layers in the architecture is not modified and directly fed to a classifier trained on the target dataset [13]. In this case, the entire pre-trained model or a massive piece of it would be trained directly on the target dataset (maybe many epochs) in order to allow all of the learned features to be adjusted according to the specific aspects of the new task [3].

C. Model Architecture

We implemented three architectures: MobileNetV2, ResNet50, and EfficientNetB0, where ResNet50 outperformed, as shown in Fig. 4.

The final layers are customised for the target classification task. Transfer learning includes fine-tuning the final layers of a pre-trained model to train it specifically for a target classification task.

Freeze and unfreeze layers in a pre-trained model during transfer learning. Offers more fine-tuned control over which layers are updated and some implicit regularisation to the model, adapting to the target task. The freezing type, on the one hand, is to prevent learned knowledge of the pre-training dataset from being overwritten (i.e. frozen), and on the other hand, is to flip this bit back in order for the fine-tuning mechanism to be able to modify this weight-sharing model according to the target dataset [3].

D. Hyperparameters Used

We tune the learning rate and batch size in experiments to optimise model performance and convergence. The learning rate of the Adam optimiser was set to 0.001 [15-17] when training the model. The pre-training model was trained for a batch size of 16 and a maximum of 50 training epochs [3].

E. Training Validation Split

After splitting the dataset into the training set and validation set in order to check how good our model generalises (and to avoid overfitting), the training set is for fitting the model, and the validation set is used to test how well our model will generalise in data it had not seen yet, with a common 80/20 split ratio.

IV. RESULTS AND ANALYSIS

A. Transfer Learning Performance

For the base pre-trained model, stable learning patterns are observed in the training and validation accuracy/loss curves in Fig. 1. (Fig1. (a) for MobileNetV2, Fig1. (b) for ResNet50 and Fig1. (c) for EfficientNetB0) The training and validation accuracies improve slowly over 10 epochs; however, the validation accuracy consistently outperforms the training accuracy, demonstrating excellent generalisation without further fine-tuning. The model achieves a validation accuracy of approximately 67.9%, with a corresponding training accuracy of 67.3% in case of MobileNetV2, other architecture results given in fig.1. Loss curves indicate convergence, with the validation loss decreasing more rapidly than the training loss, demonstrating the effective reuse of features from the pre-trained model.

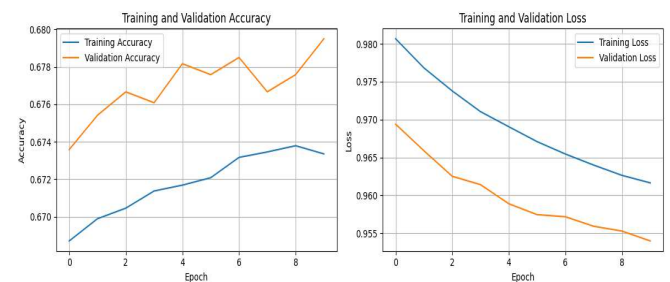


Fig.1. (a) Training and validation accuracy, Training and validation Loss for MobileNetV2 before fine-tuning.

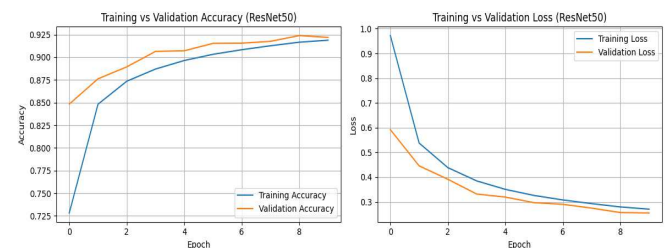


Fig.1. (b) Training and validation accuracy, Training and validation Loss for ResNet50 before fine-tune

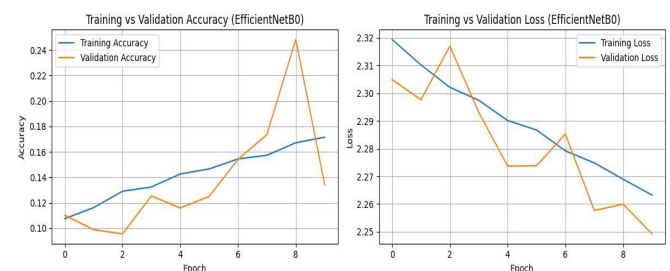


Fig.1. (c) Training and validation accuracy, Training and validation Loss for EfficientNetB0 before fine-tune

B. Fine-Tuning Enhancement

Finally, when unfreezing at testing time (as shown in Fig 2), we see a significant performance improvement. Validation accuracy increases dramatically from 25% to 88% after approximately 5 epochs, which is because the fine-tuned model can further adjust to the dataset. Also note that during the process, the training accuracy rapidly climbs to 80%, while both the training/val loss converge near 0.5

after a sharp drop, indicating that the model has enhanced its discriminatory power, specifically in learning target domain features. In Fig.2. (Fig.2. (a) for MobileNetV2, Fig.2. (b) for ResNet50 and Fig.2. (c) for EfficientNetB0) shows result for all models.

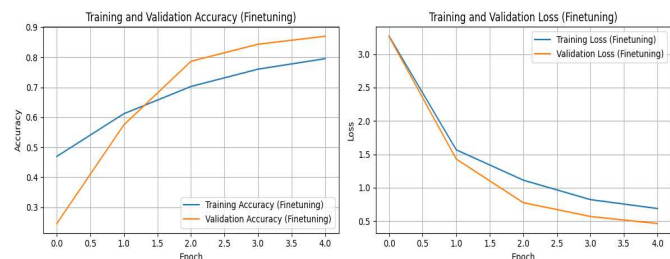


Fig.2. (a) Training and validation accuracy, Training and validation Loss for MobileNetV2 after fine-tuning.

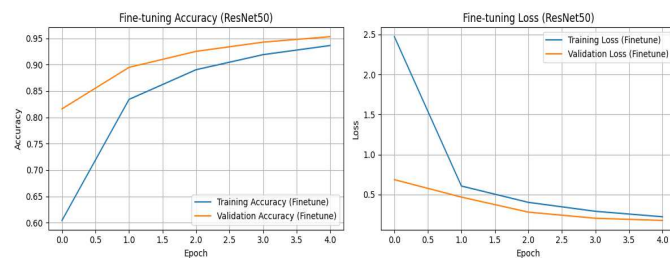


Fig.2. (b) Training and validation accuracy, Training and validation Loss for ResNet50 after fine-tuning

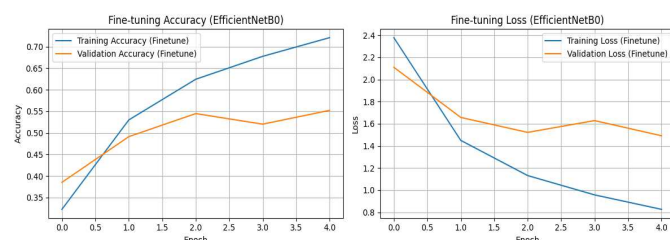


Fig.2. (c) Training and validation accuracy, Training and validation Loss for EfficientNetB0 after fine-tuning

C. Confusion Matrix Evaluation

A confusion matrix Fig.3 (Fig2. (a) for MobileNetV2, Fig2. (b) for ResNet50 and Fig2. (c) for EfficientNetB0) shows result for all models.) is used to evaluate the prediction performance with respect to each individual class. The diagonal dominance in the matrix means that there are many correct predictions for all classes.

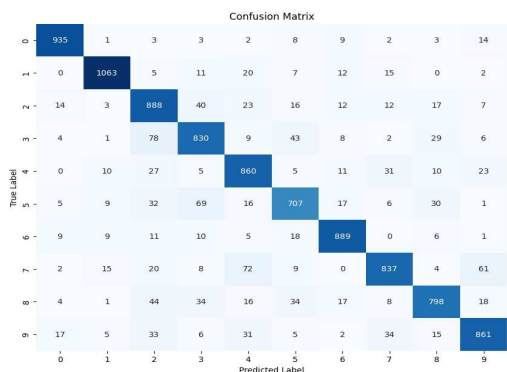


Fig.3. (a) Confusion Matrix for MobileNetV2

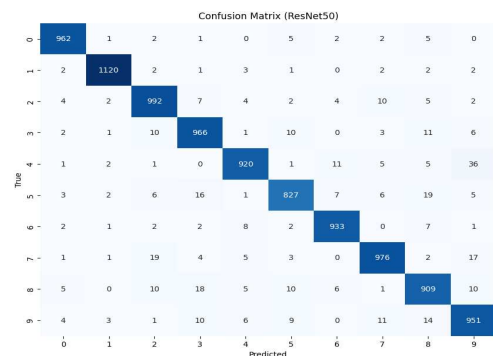


Fig.3. (b) Confusion Matrix for ResNet50

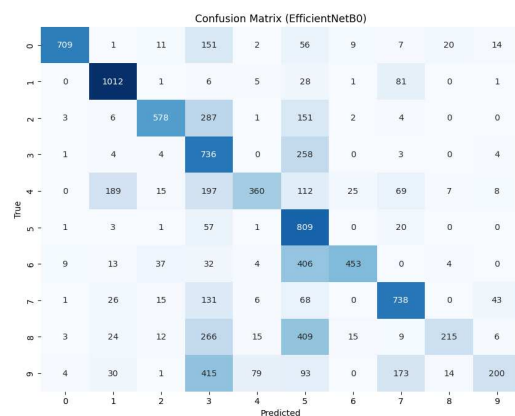


Fig.3. (c) Confusion Matrix for EfficientNetB0

The following comparison, shown in Fig. 4, presents an analysis of all three models, MobileNetV2, ResNet50, and EfficientNetB0 and their results in the form of accuracy.

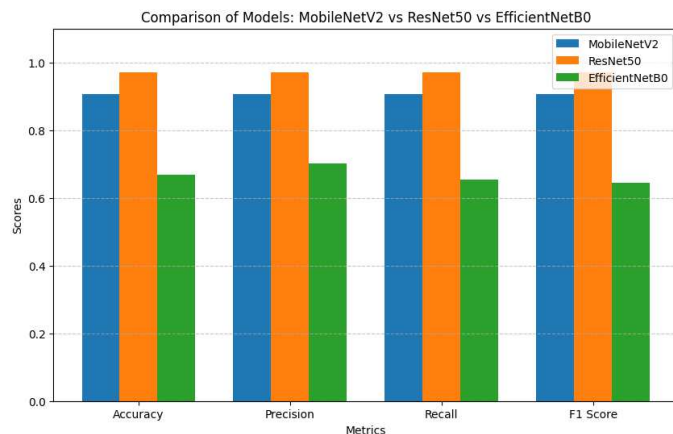


Fig.4. Comparisons for MobileNetV2, ResNet50 and EfficientNetB0 on accuracy.

D. Sample Prediction Analysis

These visualisations are invaluable for verifying model adherence and identifying failure cases, which will be something addressed in the future through uncertainty modelling or attention mechanisms. Fig.5. shows the sample classification result of MobileNetV2.

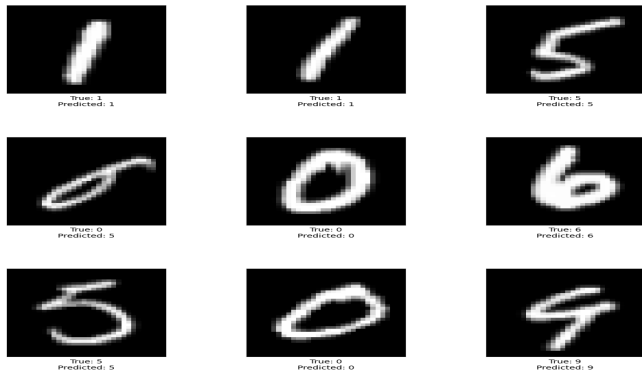


Fig.5. Some Results of MobileNetV2

V. DISCUSSION

The results obtained using transfer learning and fine-tuning are quite revealing in terms of the lessons learned about the effectiveness/utility of pre-trained models on image classification tasks, such as the handwritten digit dataset with variations.

The confusion matrix also demonstrated the model's power and shortcomings. Although most of the predictions were close to a correct one, it can be seen from Fig.4 that several digits, such as 5 and 3, or 0 and 6, led to confusion. This behaviour is common in digit recognition problems, as there are variations among the writing styles of each person. (Misclassifications) These patterns suggest that model robustness may be improved by adding more input samples, particularly from the underperforming classes, or through data augmentation.

VI. CONCLUSION

In this study, transfer learning with pre-trained convolutional neural networks was shown to be effective for image classification tasks. The first attempt with a frozen model gave mediocre performance, indicating that general visual features can be transferred across domains. However, fine-tuning made a significant difference, highlighting the importance of enabling deep learning models to learn about domain-specific idiosyncrasies.

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